

ECN 594: Price Discrimination and Self-Selection

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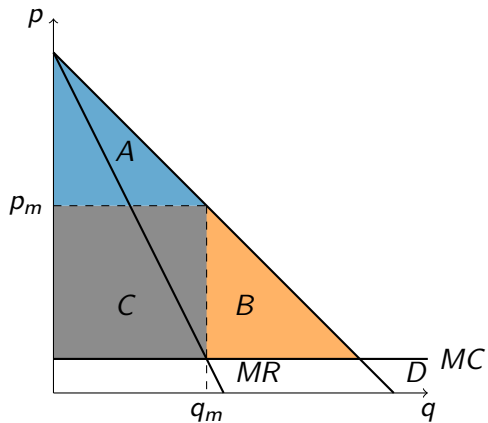
Plan

1. Types of price discrimination
2. Selection by indicators
3. Self-selection: versioning and menu design
4. Incentive compatibility and individual rationality
5. Worked example: optimal menu
6. Damaged goods examples

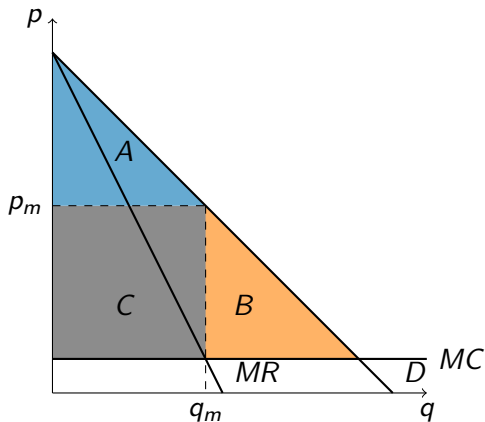
Price Discrimination

- Price discrimination: **setting different prices for the same good**
- Examples: airline tickets, software, pharmaceuticals, student discounts
- We will look at different ways firms price discriminate

Why price discriminate?



Why price discriminate?



- **Area A:** Consumers WTP $> p_m$
 - Could charge them more!
- **Area B:** Consumers WTP between MC and p_m
 - Could sell to them at lower price
- **Area C:** Current profit

Types of price discrimination (Cabral terminology)

1. Perfect price discrimination

- Charge each consumer their exact WTP
- Extracts all surplus; unrealistic benchmark

2. Selection by indicators

- Divide buyers into groups by observable characteristics
- Set different price for each group

3. Self-selection

- Cannot observe type directly
- Design menu to induce consumers to reveal type
- (Covered today)

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Selection by indicators

- Divide buyers into groups based on **observable characteristics**
- Set different price for each group
- Examples:
 - Student discounts (show student ID)
 - Senior discounts
 - Geographic pricing (different prices in different countries)
 - Time-based pricing (matinees vs evening shows)

Real-world examples of selection by indicators

1. Geographic:

- New car prices differ by region (Arizona vs California)
- Software priced differently in US vs India

2. Age-based:

- Senior discounts (more elastic, retired, fixed income)
- Student discounts (more elastic, lower income)

3. Time-based:

- Happy hour (flexible drinkers are more elastic)
- Black Friday (patient shoppers are more elastic)

Selection by indicators: setup

- Two markets: market 1 and market 2
- Demand: $q_1 = D_1(p_1)$ and $q_2 = D_2(p_2)$
- Cost: $C(q_1 + q_2)$, with constant MC
- **Goal:** Find optimal price in each market

Selection by indicators: solution

- Apply optimal pricing rule in each market:

$$MR_1 = MC \quad \text{and} \quad MR_2 = MC$$

- Equivalently, using elasticity rule:

$$\frac{p_1 - MC}{p_1} = \frac{1}{|\varepsilon_1|} \quad \text{and} \quad \frac{p_2 - MC}{p_2} = \frac{1}{|\varepsilon_2|}$$

Worked example: Optimal pricing across markets

- **Question:**
- Two markets with $\varepsilon_1 = -2$ and $\varepsilon_2 = -4$
- $MC = 6$
- Find optimal prices in each market. Should you charge a higher price in the more inelastic market?

Worked example: Optimal pricing (solution)

- **Solution:**

- Using Lerner index: $\frac{p-MC}{p} = \frac{1}{|\varepsilon|}$

- **Market 1** ($\varepsilon_1 = -2$):

$$\frac{p_1 - 6}{p_1} = \frac{1}{2} \Rightarrow p_1 - 6 = 0.5p_1 \Rightarrow p_1 = 12$$

- **Market 2** ($\varepsilon_2 = -4$):

$$\frac{p_2 - 6}{p_2} = \frac{1}{4} \Rightarrow p_2 - 6 = 0.25p_2 \Rightarrow p_2 = 8$$

- Price is higher in more inelastic market (market 1). This is a general result.

Connection: Demand estimation and price discrimination

- **Demand estimation gives us:**
 - Elasticities by consumer group (if demographics used)
 - Which groups are more/less price-sensitive
- **This informs pricing strategy:**
 - High elasticity groups $\rightarrow$ lower price
 - Low elasticity groups $\rightarrow$ higher price
- **Example:** Nevo (2001) found that families with children are less price-sensitive for kid cereals
- Cereal companies can use this for targeted promotions

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The self-selection problem: why can't we just ask?

- **Why not ask consumers their WTP directly?**
- Everyone would claim to be a low-value buyer
- **Information asymmetry:**
 - Consumers know their type
 - Firm does not
 - Consumers have incentive to misreport
- **Solution:** Design options so that revealing type is in consumers' interest

Self-selection: the idea

- **Problem:** Seller cannot directly observe consumer types
 - Can't tell who has high vs low willingness to pay
- **Solution:** Design a **menu** of options
 - Different options appeal to different types
 - Consumers “self-select” into revealing their type
- Also called: screening, menu design, second-degree price discrimination

Self-selection: versioning

- **Versioning:** Offer different “versions” of a product
 - High-quality version for high-value consumers
 - Low-quality version for low-value consumers
- **Examples:**
 - Airline tickets: business class vs economy
 - Software: Pro vs Basic editions
 - iPhone Pro vs iPhone
- **Damaged goods:** Sometimes firms deliberately reduce quality
 - Tesla: software-limited battery range
 - 19th century French rail: roofless third-class cars

The menu design problem: setup

- Two consumer types: H (high-value) and L (low-value)
- Two product versions: Full and Stripped-down
- Willingness to pay:

	Full	Stripped-down
Type H	v_H^F	v_H^S
Type L	v_L^F	v_L^S

- Assume: $v_H^F > v_L^F$ and $v_H^S > v_L^S$ (H values both more)
- Also: $v_H^F - v_H^S > v_L^F - v_L^S$ (H values upgrade more)

Consumer choice: self-selection

- Consumer surplus from each option:

$$CS_{\text{Full}} = v^F - p_F$$

$$CS_{\text{Stripped}} = v^S - p_S$$

- Each consumer chooses the option with highest CS
- (If both negative, consumer doesn't buy)

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Constraints on pricing

- **Incentive Compatibility (IC):** Each type prefers “their” option

$$IC_H : \quad v_H^F - p_F \geq v_H^S - p_S$$

$$IC_L : \quad v_L^S - p_S \geq v_L^F - p_F$$

- **Individual Rationality (IR):** Each type is willing to participate

$$IR_H : \quad v_H^F - p_F \geq 0$$

$$IR_L : \quad v_L^S - p_S \geq 0$$

- Also called: participation constraints

Key insight: which constraints bind?

- In the optimal menu:
 - IC_H binds: H is just indifferent between Full and Stripped
 - IR_L binds: L gets zero surplus from Stripped
- **Intuition:**
 - Extract all surplus from L (low type has no “outside option”)
 - H gets some surplus (to prevent them from mimicking L)
- This is called: “no distortion at the top”

Proof: which constraints bind

- **Claim 1: IR_L binds.**

- Suppose IR_L slack: $v_L^S - p_S > 0$
- Then raise p_S slightly $\rightarrow$ profit increases, L still buys
- Contradiction. So IR_L binds: $p_S = v_L^S$

- **Claim 2: IC_H binds.**

- Suppose IC_H slack: $v_H^F - p_F > v_H^S - p_S$
- Then raise p_F slightly $\rightarrow$ profit increases, H still prefers Full
- Contradiction. So IC_H binds: $v_H^F - p_F = v_H^S - p_S$

Proof: why we can ignore other constraints

- **Claim 3: IR_H is implied (doesn't bind).**

- From IC_H : $v_H^F - p_F = v_H^S - p_S$
- From IR_L : $p_S = v_L^S$, and $v_H^S > v_L^S$ (H values more)
- So $v_H^F - p_F = v_H^S - v_L^S > 0$

- **Claim 4: IC_L is implied (doesn't bind).**

- Need: $v_L^S - p_S \geq v_L^F - p_F$
- From binding constraints: $p_F - p_S = v_H^F - v_H^S$
- So need: $v_H^F - v_H^S \geq v_L^F - v_L^S$
- This is our assumption: H values the upgrade more

- **Result:** $p_S = v_L^S$ and $p_F = v_H^F - v_H^S + v_L^S$

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Worked example: Airline pricing

- **Question:**
- Business travelers (B): WTP \$800 for flexible, \$400 for restricted
- Leisure travelers (L): WTP \$300 for flexible, \$250 for restricted
- 100 of each type. $MC = 100$ for both ticket types.
- Design the optimal menu (prices for flexible and restricted).

Worked example: Airline pricing (solution)

- **Step 1:** Set IR_L to bind: $p_R = 250$ (L gets zero surplus)
- **Step 2:** Set IC_B to bind:

$$\begin{aligned}800 - p_F &= 400 - 250 \\p_F &= 800 - 150 = 650\end{aligned}$$

- Check L doesn't want flexible: $300 - 650 = -350 < 0$
- **Profit:**

$$\pi = (650 - 100) \times 100 + (250 - 100) \times 100 = 55,000 + 15,000 = 70,000$$

- B gets rent of \$150 (to prevent mimicking L)

Self-selection: versioning

- An extreme form of versioning: **damaged goods** - reduce the quality of existing products
- **Example:**



Figure: 2017 Tesla Model S full range:
\$76 thousand



Figure: Exactly the same car with a few
extra lines of code to restrict battery:
\$70 thousand

- Why would it be profitable for a seller to intentionally make some of its products worse?
Price discrimination.

Self-selection: versioning

- Another example of damaged goods (from textbook)
- 19th century French railcars: how to prevent wealthy passengers from choosing third-class tickets rather than second-class tickets?

Self-selection: versioning

- Another example of damaged goods (from textbook)
- 19th century French railcars: how to prevent wealthy passengers from choosing third-class tickets rather than second-class tickets?
- Answer: pull the roof off the third-class railcar!

Worked example: Versioning

- Two product versions. $MC = 300$ for both.
- 1 million type H consumers; 2 million type L consumers
- Willingness to pay:

	Full	Stripped
Type H	1500	800
Type L	600	500

- **Questions:**
- (a) Profit from selling only Full at $p_F = 1500$?
- (b) Profit from $p_F = 1500$, $p_S = 500$?
- (c) Profit from $p_F = 1200$, $p_S = 500$?

Worked example: Versioning (solution part a)

- **(a) Only Full at $p_F = 1500$:**
 - Type H: $CS = 1500 - 1500 = 0 \geq 0$, buys
 - Type L: $CS = 600 - 1500 = -900 < 0$ (doesn't buy)
- Only H buys
- Profit = $(1500 - 300) \times 1M = \$1.2B$

Worked example: Versioning (solution part b)

- **(b)** $p_F = 1500$, $p_S = 500$:
- Type H:
 - $CS_F = 1500 - 1500 = 0$
 - $CS_S = 800 - 500 = 300 \leftarrow$ higher
- Type H buys Stripped (IC violated)
- Type L: $CS_S = 500 - 500 = 0$ (buys Stripped)
- Everyone buys Stripped
- Profit = $(500 - 300) \times 3M = \$600M$
- Worse than (a). The menu is poorly designed.

Worked example: Versioning (solution part c)

- (c) $p_F = 1200$, $p_S = 500$:
- Type H:
 - $CS_F = 1500 - 1200 = 300$
 - $CS_S = 800 - 500 = 300$
- H is indifferent $\rightarrow$ buys Full (IC binds)
- Type L: $CS_F = 600 - 1200 = -600$, $CS_S = 500 - 500 = 0$
- L buys Stripped (IR binds)
- Profit = $(1200 - 300) \times 1M + (500 - 300) \times 2M = \$1.3B$
- Best of the three

Lessons from the example

- The optimal menu satisfies:
 1. IC for H binds: H just willing to buy Full over Stripped
 2. IR for L binds: L gets exactly zero surplus
- H gets “information rent” (surplus of 300)
 - This is the cost of preventing H from mimicking L
- Price gap $p_F - p_S$ must be $\leq v_H^F - v_H^S$ (otherwise H switches)

Self-selection: cookbook summary

1. Compute consumer surplus for each type, for each option:

$$CS = WTP - price$$

2. For each type, find the option with highest CS
3. Check that $CS \geq 0$ (otherwise they don't buy)
4. Compute profit given consumer choices
5. To find optimal prices: set IC for H and IR for L to bind

Pricing/Price discrimination

Type	Key requirement	Examples
Selection by indicators	Observable characteristic	Students, seniors, geographic regions
Two-part tariffs	Can charge fixed + variable	Gym, Costco, phone plans
Self-selection (versioning)	Can offer different versions	Airline classes, software editions

- All require: **market power** and **inability to resell**

Connection to demand estimation

- Demand estimation gives us **elasticities by consumer type**
- This directly informs pricing strategy:
 - **Selection by indicators:** Different prices for observable groups
 - **Versioning:** Design features that appeal differently to types
- **Example:** If demand estimation shows:
 - Business travelers inelastic, value flexibility
 - Leisure travelers elastic, value low price
- $\Rightarrow$ Offer flexible tickets at high price, restricted at low price

Key Points

1. **Selection by indicators:** Observable characteristics, apply Lerner index to each group
2. Higher prices in more inelastic markets
3. **Self-selection:** Design menu so types reveal themselves through their choices
4. **IC constraint:** Each type prefers “their” option
5. **IR constraint:** Each type willing to participate ($CS \geq 0$)
6. Optimal menu: IC binds for high type, IR binds for low type
7. High type gets “information rent”; low type gets zero surplus
8. **Damaged goods:** Firms may deliberately reduce quality to segment market